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Clinical Study

Gamma knife stereotactic radiosurgery yields good long-term outcomes for low-volume uveal melanomas without intraocular complications

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ABSTRACT

We present the outcomes of 35 uveal melanoma patients treated with gamma knife stereotactic radiosurgery. All cases were previously untreated. During follow-up, regular MRI examinations were used to detect any changes in tumor size and estimate the local long-term tumor control rate. Treatment-related complications were also recorded. During follow-up, systemic dissemination was observed in two patients, one of whom died of metastases. The most frequent complication was retinal detachment (17.1%). Three patients required enucleation. Cumulative 1-year and 3-year local tumor growth control rates were 97% and 83%, respectively. The mean and median times to local tumor progression were 48.0 and 51.7 months, respectively. Gamma knife surgery may be a suitable alternative for the treatment of low-volume uveal tumors without intraocular complications, as the control rate and long-term outcomes compare favorably with those of surgical excision and brachytherapy.

1. Introduction

Uveal melanoma is the most common primary ocular malignancy in adults, with an annual incidence of 6–7 cases per million people.[1–3] The choroid is the most common site of involvement, followed by the ciliary body and iris, the latter being least frequently affected.[4] The disease is most commonly diagnosed in the sixth decade of life, and the incidence of this tumor rises dramatically with age.[5] The determinants of mortality for malignant melanoma of the choroid include tumor size and presence of systemic dissemination as well as features such as location, tumor height, growth rate and degree of tumor pigmentation.[4]

The treatment modalities for choroidal melanoma include observation, photocoagulation, transpupillary thermotherapy, radiotherapy, local resection, orbital exenteration, chemotherapy, immunotherapy, enucleation, and stereotactic radio surgery, depending on the clinical condition of the patient. Of these, photocoagulation and transpupillary thermotherapy are reserved for the management of selected cases with very small tumors.[6–8] For posterior uveal melanomas, radiotherapy remains the most widely used method of treatment.[9–11] Enucleation is indicated for advanced melanomas that occupy most intraocular structures, that have caused severe secondary glaucoma, and that have invaded the optic nerve.[5] The principal objective of the treatment is to reduce the risk of death from metastasis and to conserve as much vision as possible while minimizing the risk of enucleation.

Here, we present our experience with gamma knife stereotactic radiosurgery for treating uveal melanomas. We also present an analysis of the outcomes and radiation-induced complications associated with this surgery.

2. Materials and methods

2.1. Patients

During a 7-year period between January 2001 and January 2008, 35 patients with uveal melanoma underwent gamma knife radiosurgery (GKS) at the Gamma Knife Unit of Marmara University Hospital. The patient population comprised 16 women and 19 men, with a mean age of 57.3 years (range: 26–81 years). Exclusion criteria included prior treatment and presence of metastases at the time of diagnosis as confirmed by positron emission tomography. The diagnosis of uveal melanoma was based on radiological and fundoscopic findings. All patients had been diagnosed and referred by ophthalmology clinics and had undergone an orbital examination preoperatively. Before GKS, baseline measurements were

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